

Language Transfer Guidance

The Model of Programming Language Transfer¹ is used to transition pupils from Scratch to Python.

TCC = True Carryover Concept

- Pupils have already seen this construct in Scratch. We can teach it in Python assuming they have the correct idea of how it works. An example of this is penup() in Python. This works the same as using the penup block in Scratch.

ATCC = Abstract True Carryover Concept

- Pupils have seen this construct in Scratch but it *looks* different in Python. We should make explicit comparisons between the Scratch and Python construct to make sure pupils understand that it is the same idea... it just looks different! An example of this is the print() function in Python. This does the same thing as the say block in Scratch even though the command word is different.

FCC = False Carryover Concept

- Pupils have seen a construct in Scratch that *looks* similar but is a different concept to what we are teaching in Python. We should be very clear in explaining what the differences are between the two constructs to make sure pupils do not make false assumptions. In Scratch, it is not possible to draw a circle so pupils might falsely assume that this command will not work in Python.

Guess Quiz Guide and Answer Key

Question no.	Concept(s)	Classification	Quiz Answer
1	Printing	ATCC	A
2	Variables	ATCC	B
3	If/else construct	ATCC	D
4	For loops	ATCC (possible FCC due to differences in loop terminations)	C
5	Turtle (forward and color)	ATCC	C
6	Turtle (penup/down + forward + color)	TCC (penup/down), ATCC (penup/down + color)	A

¹ Tshukudu, E., Cutts, Q. and Foster, M.E. (2021) 'Evaluating a Pedagogy for Improving Conceptual Transfer and Understanding in a Second Programming Language Learning Context', *Proceedings of the 21st Koli Calling International Conference on Computing Education Research*. New York, NY, USA: Association for Computing Machinery (Koli Calling '21), pp. 1–10. Available at: <https://doi.org/10.1145/3488042.3488050>.

7	Turtle (left/right + forward + color + pendown)	TCC (left/right + pendown), ATCC (forward + color)	D
8	Turtle (circle)	FCC (circle does not exist in Scratch)	Draws a red circle
9	If/else construct with a variable	ATCC	Draws a green circle

Implementation in lessons

- Slides for lessons feature images of Scratch blocks for comparison where necessary.
- The guess quiz should be used to give an idea of how naturally pupils are able to connect new Python constructs to what they have learned previously in Scratch. Even if a concept is marked as TCC, if many pupils get this question wrong this is an indication that this concept may requires further explanation.